Import Libraries

[1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

Load Dataset

[2]:
```
df = pd.read_csv('Students Performance Dataset.csv')
print(df)
```

```
    Student ID Student Name  Python  Java  Linux  AI  SQL  Total Attendance
0         1001         Abhi      41    53     33  95   45    267        97%
1         1002        Abhay      37    60     78  71   67    313        98%
2         1003      Bhaskar      34    65     72  57   38    266        98%
3         1004         Bala      61    80     61  78   51    331        98%
4         1005        Bhati      71    59     65  51   59    305        98%
5         1006     Chandran      77    47     90  54   83    351        98%
6         1007       Chawla      97    92     82  72   75    418        98%
7         1008         Dora      93    42     86  51   67    339        98%
8         1009     Francies      39    90     65  35   46    275       100%
9         1010       Gerage      58    36     88  41   61    284        96%
10        1011        Gopal      55    48     77  37   96    313        99%
11        1012     JaiKanth      48    60     54  84   52    298        94%
12        1013       Karthi      99    76     48  60   98    381        98%
13        1014       Kandan      64    92     85  37   61    339        96%
14        1015         Kala      44    66     90  84   93    377        98%
15        1016        Kamal      77    75     63  39   52    306       100%
16        1017          Lal      94    69     71  96   91    421       100%
17        1018        Lalli      47    72     42  84   60    305       100%
```

[3]:
```
df.head()
```

[3]:

	Student ID	Student Name	Python	Java	Linux	AI	SQL	Total	Attendance
0	1001	Abhi	41	53	33	95	45	267	97%
1	1002	Abhay	37	60	78	71	67	313	98%
2	1003	Bhaskar	34	65	72	57	38	266	98%
3	1004	Bala	61	80	61	78	51	331	98%
4	1005	Bhati	71	59	65	51	59	305	98%

[4]:
```
df.tail()
```

[4]:

	Student ID	Student Name	Python	Java	Linux	AI	SQL	Total	Attendance
13	1014	Kandan	64	92	85	37	61	339	96%
14	1015	Kala	44	66	90	84	93	377	98%
15	1016	Kamal	77	75	63	39	52	306	100%
16	1017	Lal	94	69	71	96	91	421	100%
17	1018	Lalli	47	72	42	84	60	305	100%

[5]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 18 entries, 0 to 17
Data columns (total 9 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Student ID    18 non-null     int64
 1   Student Name  18 non-null     object
 2   Python        18 non-null     int64
 3   Java          18 non-null     int64
 4   Linux         18 non-null     int64
 5   AI            18 non-null     int64
 6   SQL           18 non-null     int64
 7   Total         18 non-null     int64
 8   Attendance    18 non-null     object
dtypes: int64(7), object(2)
memory usage: 1.4+ KB
```

[6]:
```
df.describe()
```

[6]:

	Student ID	Python	Java	Linux	AI	SQL	Total
count	18.000000	18.000000	18.000000	18.000000	18.000000	18.000000	18.000000
mean	1009.500000	63.111111	65.666667	69.444444	62.555556	66.388889	327.166667
std	5.338539	22.071778	16.803361	17.037061	20.881239	18.824699	47.234397
min	1001.000000	34.000000	36.000000	33.000000	35.000000	38.000000	266.000000
25%	1005.250000	44.750000	54.500000	61.500000	43.500000	52.000000	299.750000
50%	1009.500000	59.500000	65.500000	71.500000	58.500000	61.000000	313.000000
75%	1013.750000	77.000000	75.750000	84.250000	82.500000	81.000000	348.000000
max	1018.000000	99.000000	92.000000	90.000000	96.000000	98.000000	421.000000

[7]:
```
df.isnull().sum()
```

[7]:
```
Student ID      0
Student Name    0
Python          0
Java            0
Linux           0
AI              0
SQL             0
Total           0
Attendance      0
dtype: int64
```

[9]:
```
df.duplicated().sum()
```

[9]:
```
np.int64(0)
```

[21]:
```
import pandas as pd

df = pd.read_csv('Students Performance Dataset.csv')
df.columns = df.columns.str.strip()

SUBJECTS = ["Python", "Java", "Linux", "AI", "SQL"]
DEPARTMENTS = ["Computer Science", "Information Technology", "AI and Data Science"]

df["Attendance"] = pd.to_numeric(
    df["Attendance"].astype(str).str.rstrip("%"), errors="coerce"
)

df["Department"] = [DEPARTMENTS[i % len(DEPARTMENTS)] for i in range(len(df))]

df = df.rename(columns={"Total": "Source_Total"})
df = df[["Student ID", "Student Name", "Department"] + SUBJECTS + ["Attendance", "Source_Total"]]

df.head()
```

[21]:

	Student ID	Student Name	Department	Python	Java	Linux	AI	SQL	Attendance	Source_Total
0	1001	Abhi	Computer Science	41	53	33	95	45	97	267
1	1002	Abhay	Information Technology	37	60	78	71	67	98	313
2	1003	Bhaskar	AI and Data Science	34	65	72	57	38	98	266
3	1004	Bala	Computer Science	61	80	61	78	51	98	331
4	1005	Bhati	Information Technology	71	59	65	51	59	98	305

Rules Used in the Student Performance System

Grading Rule (based on the student's ** average marks** ; every subject is out of 100)

Average marks	Grade
90 and above	**A+**
80 - 89.99	**A**
70 - 79.99	**B**
60 - 69.99	**C**
50 - 59.99	**D**
Below 50	**F**

Pass / Fail rule

A student **passes** only if **both** conditions hold:

1. Marks ≥ **40** in **every** subject, and
2. Attendance ≥ **75%**.

Otherwise the student **fails**. (Change the constants below to use a different rule.)

[23]:
```
MAX_MARKS_PER_SUBJECT = 100
PASS_MARK = 40
MIN_ATTENDANCE = 75
TOP_N = 5
GRADE_BOUNDARIES = [(90,"A+"),(80,"A"),(70,"B"),(60,"C"),(50,"D")]
LOWEST_GRADE = "F"
```

Reusable Functions

[35]:
```python
import numpy as np

SUBJECTS = ["Python", "Java", "Linux", "AI", "SQL"]
PASS_MARK = 40
MIN_ATTENDANCE = 75
GRADE_BOUNDARIES = [(90, "A+"), (80, "A"), (70, "B"), (60, "C"), (50, "D")]
LOWEST_GRADE = "F"

def calculate_total(marks):
    return np.sum(marks, axis=1)

def calculate_average(marks):
    return np.round(np.mean(marks, axis=1), 2)

def assign_grade(average):
    for lower_bound, grade in GRADE_BOUNDARIES:
        if average >= lower_bound:
            return grade
    return LOWEST_GRADE

def check_pass_fail(marks_row, attendance):
    passed_all_subjects = np.all(marks_row >= PASS_MARK)
    if passed_all_subjects and attendance >= MIN_ATTENDANCE:
        return "Pass"
    return "Fail"

def failed_subjects(marks_row, subjects=SUBJECTS):
    return [subj for subj, m in zip(subjects, marks_row) if m < PASS_MARK]

def subject_statistics(column):
    column = np.asarray(column)
    return {
        "Average": round(float(np.mean(column)), 2),
        "Median": float(np.median(column)),
        "Highest": int(np.max(column)),
        "Lowest": int(np.min(column)),
        "Std Deviation": round(float(np.std(column)), 2),
        "Pass %": round(float(np.mean(column >= PASS_MARK) * 100), 2),
    }
```

Student-wise Results

Total, average, grade and pass/fail for each student.

[39]:
```
MAX_MARKS_PER_SUBJECT = 100

df = df.reset_index(drop=True)

marks = df[SUBJECTS].to_numpy()

df["Total"] = calculate_total(marks)
df["Average"] = calculate_average(marks)
df["Percentage"] = np.round(
    df["Total"] / (len(SUBJECTS) * MAX_MARKS_PER_SUBJECT) * 100, 2
)

grades, statuses, fails = [], [], []
for idx in range(len(df)):
    grades.append(assign_grade(df.loc[idx, "Average"]))
    statuses.append(check_pass_fail(marks[idx], df.loc[idx, "Attendance"]))
    fails.append(",".join(failed_subjects(marks[idx])) or "-")

df["Grade"] = grades
df["Status"] = statuses
df["Failed Subjects"] = fails

mismatches = int((df["Total"] != df["Source_Total"]).sum())
print(f"Rows where calculated total != file total: {mismatches}")

df[["Student ID", "Student Name", "Department", "Total", "Average",
    "Grade", "Status", "Failed Subjects"]]
```

```
Rows where calculated total != file total: 0
```

[39]:

	Student ID	Student Name	Department	Total	Average	Grade	Status	Failed Subjects
0	1001	Abhi	Computer Science	267	53.4	D	Fail	Linux
1	1002	Abhay	Information Technology	313	62.6	C	Fail	Python
2	1003	Bhaskar	AI and Data Science	266	53.2	D	Fail	Python,SQL
3	1004	Bala	Computer Science	331	66.2	C	Pass	-
4	1005	Bhati	Information Technology	305	61.0	C	Pass	-
5	1006	Chandran	AI and Data Science	351	70.2	B	Pass	-
6	1007	Chawla	Computer Science	418	83.6	A	Pass	-
7	1008	Dora	Information Technology	339	67.8	C	Pass	-
8	1009	Francies	AI and Data Science	275	55.0	D	Fail	Python,AI
9	1010	Gerage	Computer Science	284	56.8	D	Fail	Java
10	1011	Gopal	Information Technology	313	62.6	C	Fail	AI
11	1012	JaiKanth	AI and Data Science	298	59.6	D	Pass	-
12	1013	Karthi	Computer Science	381	76.2	B	Pass	-
13	1014	Kandan	Information Technology	339	67.8	C	Fail	AI
14	1015	Kala	AI and Data Science	377	75.4	B	Pass	-
15	1016	Kamal	Computer Science	306	61.2	C	Fail	AI
16	1017	Lal	Information Technology	421	84.2	A	Pass	-
17	1018	Lalli	AI and Data Science	305	61.0	C	Pass	-

Grade distribution and pass/fail summary ¶

[88]:
```
from IPython.display import display

subjects = ["Python", "SQL", "Java", "Linux", "AI"]

# Create the missing "Failed Subjects" column (subjects scored below 50)
df["Failed Subjects"] = df[subjects].apply(
    lambda row: ", ".join(row.index[row < 50]) or "None", axis=1
)

# Grade distribution
grade_order = ["A+", "A", "B", "C", "D", "F"]
grade_counts = df["Grade"].value_counts().reindex(grade_order, fill_value=0)
print("Grade distribution:")
print(grade_counts.to_string())

# Pass / fail summary
n_students = len(df)
n_pass = int((df["Result"] == "Pass").sum())
n_fail = n_students - n_pass
print(f"\nPassed: {n_pass}  |  Failed: {n_fail}  |  Pass rate: {n_pass / n_students * 100:.1f}%")

# Failed students
print("\nStudents who failed:")
failed = df.loc[df["Result"] == "Fail", ["Student_ID", "Name", "Average", "Grade", "Failed Subjects"]]

if failed.empty:
    print("No students failed.")
else:
    display(failed.reset_index(drop=True))
```

```
Grade distribution:
Grade
A+    0
A     2
B     3
C     8
D     5
F     0

Passed: 13  |  Failed: 5  |  Pass rate: 72.2%

Students who failed:
```

	Student ID	Name	Average	Grade	Failed Subjects
0	1014	JaiKanth	59.6	D	Python
1	1015	Gerage	56.8	D	Java, AI
2	1016	Francies	55.0	D	Python, SQL, AI
3	1017	Abhi	53.4	D	Python, SQL, Linux
4	1018	Bhaskar	53.2	D	Python, SQL

Overall Performance: Highest, Lowest & Average

[48]:
```
totals = df["Total"].to_numpy()
best_i = int(np.argmax(totals))
worst_i = int(np.argmin(totals))

print("-" * 55)
print("OVERALL CLASS PERFORMANCE")
print("-" * 55)
print(f"Highest total : {totals[best_i]} -> {df.loc[best_i, 'Student Name']}"
      f"Avg : {df.loc[best_i,'Average']},Grade {df.loc[best_i, 'Grade']}")
print(f"Lowest total : {totals[worst_i]} -> {df.loc[worst_i, 'Student Name']}"
      f"Avg : {df.loc[worst_i,'Average']},Grade {df.loc[worst_i, 'Grade']}")
print(f"Class average total : {np.mean(totals):2f}")
print(f"Class average marks : {np.mean(df['Average']):.2f}")
print(f"Median total        : {np.median(totals):2f}")
print(f"Standard deviation : {np.std(totals):.2f}")
print(f"Average attendance : {np.mean(df['Attendance']):.2f}%")
```

```
-------------------------------------------------------
OVERALL CLASS PERFORMANCE
-------------------------------------------------------
Highest total : 421 -> LalAvg : 84.2,Grade A
Lowest total : 266 -> BhaskarAvg : 53.2,Grade D
Class average total : 327.166667
Class average marks : 65.43
Median total        : 313.000000
Standard deviation : 45.90
Average attendance : 98.00%
```

[46]:
```
subjects_rows = {}
for subject in SUBJECTS:
    subjects_rows[subject] = subject_statistics(df[subject].to_numpy())
subject_analysis = pd.DataFrame(subjects_rows).T
toppers, weakest = [],[]
for subject in SUBJECTS:
    col = df[subject].to_numpy()
    toppers.append(f"{df.loc[np.argmax(col), 'Student Name']} ({col.max()})")
    weakest.append(f"{df.loc[np.argmax(col), 'Student Name']} ({col.min()})")
subject_analysis["Topper"] = toppers
subject_analysis["Lowest Scorer"] = weakest
subject_analysis
```

[46]:

	Average	Median	Highest	Lowest	Std Deviation	Pass %	Topper	Lowest Scorer
Python	63.11	59.5	99.0	34.0	21.45	83.33	Karthi (99	Karthi (34
Java	65.67	65.5	92.0	36.0	16.33	94.44	Chawla (92	Chawla (36
Linux	69.44	71.5	90.0	33.0	16.56	94.44	Chandran (90	Chandran (33
AI	62.56	58.5	96.0	35.0	20.29	77.78	Lal (96	Lal (35
SQL	66.39	61.0	98.0	38.0	18.29	94.44	Karthi (98	Karthi (38

[48]:
```
strongest = subject_analysis["Average"].idxmax()
weakest_subj = subject_analysis["Average"].idxmin()
hardest_pass = subject_analysis["Pass %"].idxmin()
print(f"Strongest subject (highest class average): {strongest} ({subject_analysis.loc[strongest, 'Average']})")
print(f"Weakest subject (lowest class average) :   {weakest_subj} ({subject_analysis.loc[weakest_subj, 'Average']})")
print(f"Subject with lowest pass %             : {hardest_pass} ({subject_analysis.loc[hardest_pass, 'Pass %' ]}%)")
```

```
Strongest subject (highest class average): Linux (69.44)
Weakest subject (lowest class average) :   AI (62.56)
Subject with lowest pass %             : AI (77.78%)
```

Top-performing Students

[50]:
```
df["Rank"] = df["Total"].rank(method="min", ascending=False).astype(int)
top_students = (df.sort_values(["Total","Attendance"], ascending=False)
                .head(TOP_N)[["Rank","Student ID","Student Name","Department",
                              "Total","Average","Grade","Attendance"]]
                .reset_index(drop=True))
print(f"Top {TOP_N} students")
top_students
```

```
Top 5 students
```

[50]:

	Rank	Student ID	Student Name	Department	Total	Average	Grade	Attendance
0	1	1017	Lal	Information Technology	421	84.2	A	100
1	2	1007	Chawla	Computer Science	418	83.6	A	98
2	3	1013	Karthi	Computer Science	381	76.2	B	98
3	4	1015	Kala	AI and Data Science	377	75.4	B	98
4	5	1006	Chandran	AI and Data Science	351	70.2	B	98

[51]:
```
print("A+ students:", df.loc[df["Grade"] == "A+", "Student Name"].tolist() or "None")
print("\nTop Student per department:")
for dept in DEPARTMENTS:
    sub = df[df["Department"] ==dept]
    best = sub.loc[sub["Total"].idxmax()]
    print(f" {dept:<24} -> {best['Student Name']} (Total {best['Total']}, Avg {best['Average']})")
```

```
A+ students: None

Top Student per department:
 Computer Science         -> Chawla (Total 418, Avg 83.6)
 Information Technology   -> Lal (Total 421, Avg 84.2)
 AI and Data Science      -> Kala (Total 377, Avg 75.4)
```

Department-Wise Summary

[53]:
```
dept_summary = (df.groupby("Department").agg(Students=("Student ID","count"),
                                             Avg_Marks=("Average", "mean"),
                                             Highest_Total=("Total","max"),
                                             Lowest_Total=("Total","min"),
                                             Avg_Attendance=("Attendance","mean"),
                                             Passed=("Status", lambda s: (s == "Pass").sum())).round(2))
dept_summary["Pass %"] = np.round(dept_summary["Passed"] / dept_summary["Students"] * 100 ,2)
dept_summary
```

[53]:

Department	Students	Avg Marks	Highest Total	Lowest Total	Avg Attendance	Passed	Pass %
AI and Data Science	6	62.40	377	266	98.00	4	66.67
Computer Science	6	66.23	418	267	97.83	3	50.00
Information Technology	6	67.67	421	305	98.17	3	50.00

Visualisations

[60]:
```
import matplotlib.pyplot as plt
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

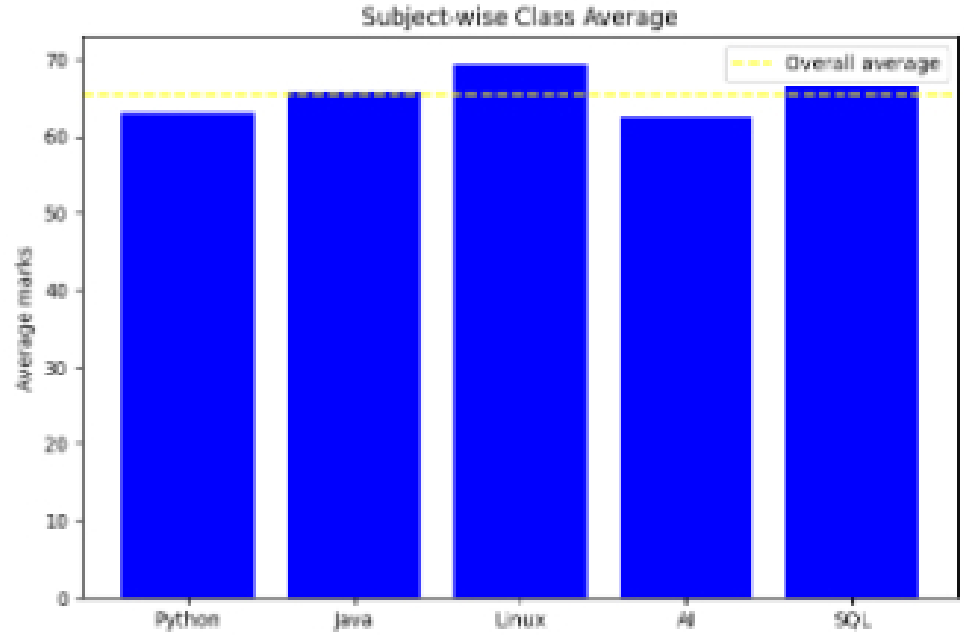
# 1) Subject-wise class average
axes[0, 0].bar(subject_analysis.index, subject_analysis["Average"], color="blue")
axes[0, 0].axhline(df["Average"].mean(), color="yellow", linestyle="--", label="Overall average")
axes[0, 0].set_title("Subject-wise Class Average")
axes[0, 0].set_ylabel("Average marks")
axes[0, 0].legend()

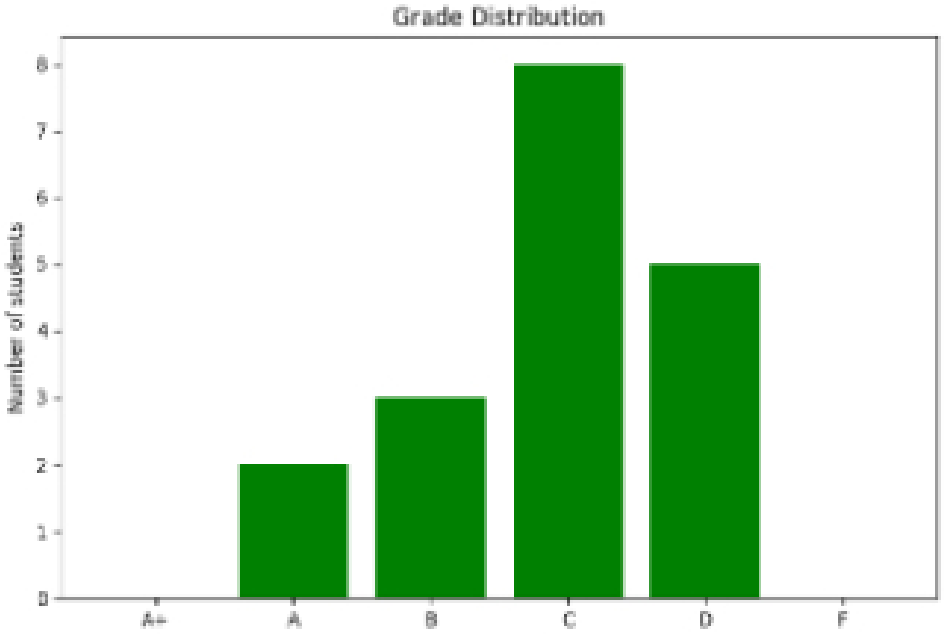
# 2) Grade distribution
grade_order = ["A+", "A", "B", "C", "D", "F"]
grade_counts = df["Grade"].value_counts().reindex(grade_order, fill_value=0)
axes[0, 1].bar(grade_counts.index, grade_counts.values, color="green")
axes[0, 1].set_title("Grade Distribution")
axes[0, 1].set_ylabel("Number of students")

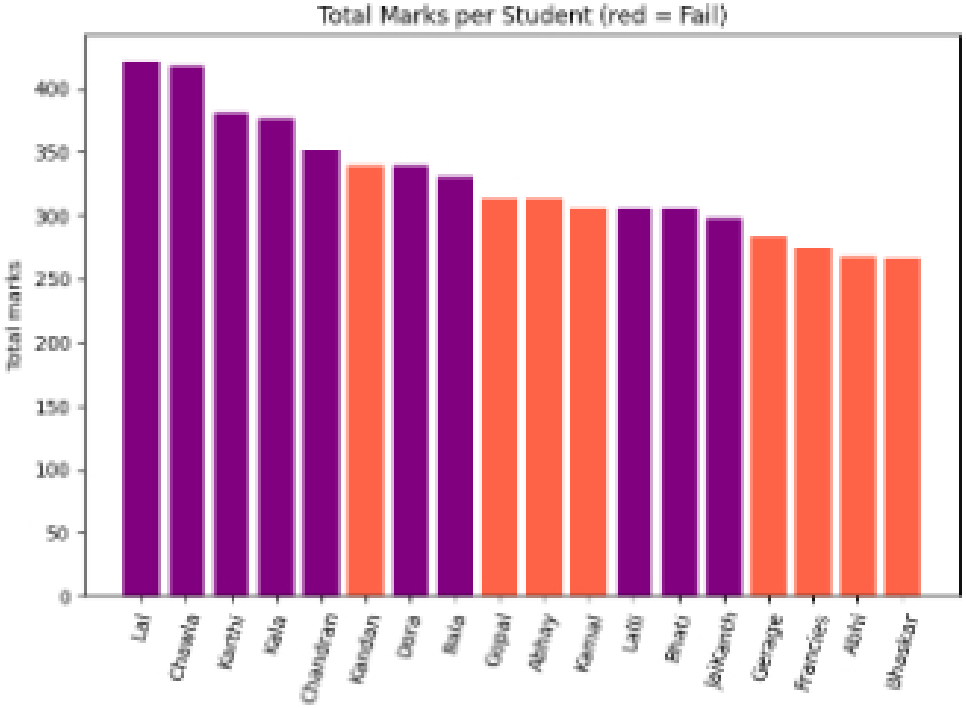
# 3) Total marks per student
order = df.sort_values("Total", ascending=False)
colors = ["tomato" if s == "Fail" else "purple" for s in order["Status"]]
axes[1, 0].bar(order["Student Name"], order["Total"], color=colors)
axes[1, 0].set_title("Total Marks per Student (red = Fail)")
axes[1, 0].tick_params(axis="x", rotation=75)
axes[1, 0].set_ylabel("Total marks")

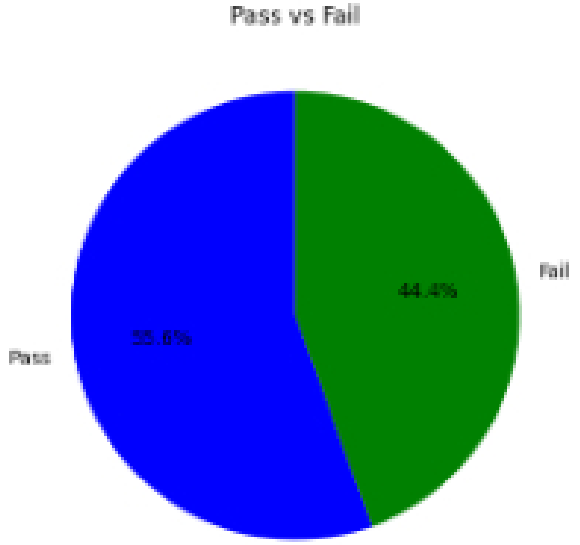
# 4) Pass vs Fail
n_pass = int((df["Status"] == "Pass").sum())
n_fail = int((df["Status"] == "Fail").sum())
axes[1, 1].pie([n_pass, n_fail], labels=["Pass", "Fail"], autopct="%1.1f%%",
               colors=["blue", "green"], startangle=90)
axes[1, 1].set_title("Pass vs Fail")

plt.tight_layout()
plt.show()
```

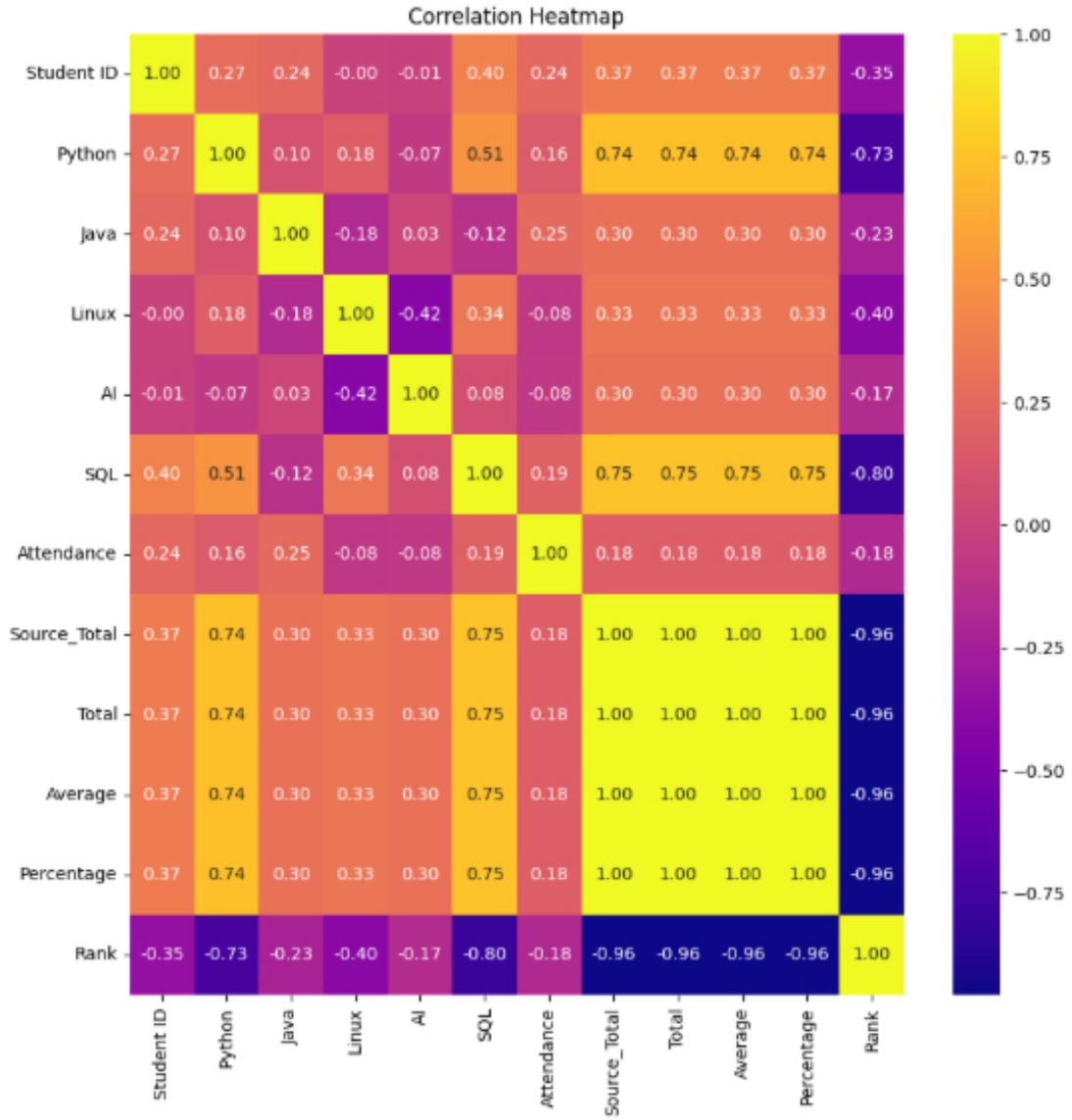

Subject-wise Class Average
Overall average
Average marks
0
10
20
30
40
50
60
70
Python
Java
Linux
AI
SQL


Grade Distribution
Number of students
0
1
2
3
4
5
6
7
8
A+
A
B
C
D
F


Total Marks per Student (red = Fail)
Total marks
0
50
100
150
200
250
300
350
400
Lal
Chawla
Karthi
Kala
Chandran
Kandan
Dora
Rata
Gopal
Abhay
Kamal
Lalli
Bhati
Jaikanth
Gerage
Francies
Abhi
Bhaskar


Pass vs Fail
Pass
55.6%
Fail
44.4%

[63]:
```
import pandas as pd
import matplotlib.pyplot as plt
plt.figure (figsize=(10,10))
correlation = df.corr(numeric_only=True)
sns.heatmap (
      correlation,
      annot=True,
      cmap='plasma',
      fmt=".2f"
)
plt.title("Correlation Heatmap")
plt.show()
```



Data Analysis Questions and Answers

Which student has the highest average?

[70]:
```
students = df.groupby("Student Name")["Total"].mean().idxmax()
print("Name of the Student with Highest Score", students)
```

```
Name of the Student with Highest Score Lal
```

Which subject has the best average?

[71]:
```
best_subject = subject_analysis["Average"].idxmax()
best_average =subject_analysis["Average"].max()
print(f"Best subject: {best_subject} (average {best_average})")
```

```
Best subject: Linux (average 69.44)
```

How many students passed?

[75]:
```
passed = (df["Status"] == "Pass").sum()
print("Number of Students Passed:",passed)
```

```
Number of Students Passed: 10
```

What is the overall class average?

[78]:
```
class_Total = df["Total"].mean()
print("Overall class average", class_Total)
```

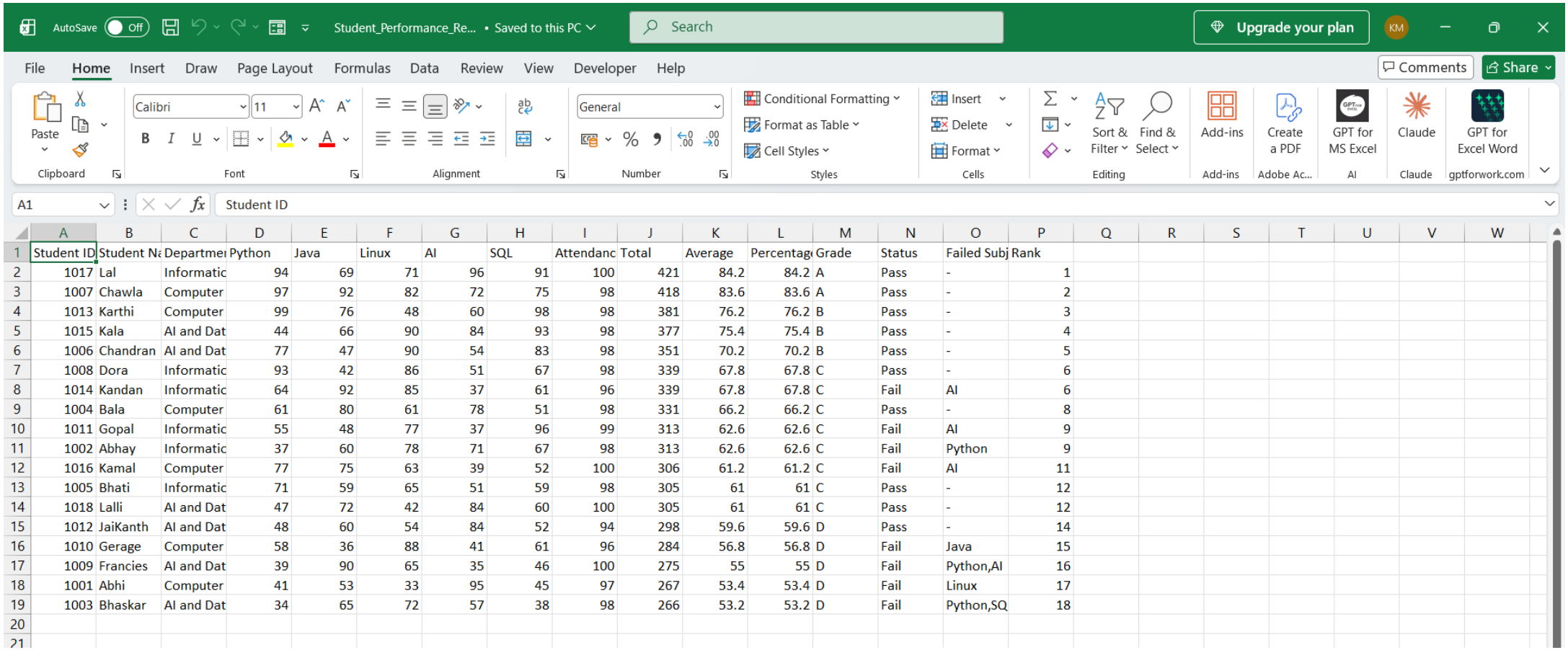
```
Overall class average 327.16666666666667
```

[64]:
```
final_report = df.drop(columns=["Source_Total"]).sort_values("Rank").reset_index(drop=True)
final_report.to_csv("Student_Performance_Report.csv", index=False)
print("Saved: Student_Performance_Report.csv")
final_report
```

```
Saved: Student_Performance_Report.csv
```

[64]:

	Student ID	Student Name	Department	Python	Java	Linux	AI	SQL	Attendance	Total	Average	Percentage	Grade	Status	Failed Subjects	Rank
0	1017	Lal	Information Technology	94	69	71	96	91	100	421	84.2	84.2	A	Pass	-	1
1	1007	Chawla	Computer Science	97	92	82	72	75	98	418	83.6	83.6	A	Pass	-	2
2	1013	Karthi	Computer Science	99	76	48	60	98	98	381	76.2	76.2	B	Pass	-	3
3	1015	Kala	AI and Data Science	44	66	90	84	93	98	377	75.4	75.4	B	Pass	-	4
4	1006	Chandran	AI and Data Science	77	47	90	54	83	98	351	70.2	70.2	B	Pass	-	5
5	1008	Dora	Information Technology	93	42	86	51	67	98	339	67.8	67.8	C	Pass	-	6
6	1014	Kandan	Information Technology	64	92	85	37	61	96	339	67.8	67.8	C	Fail	AI	6
7	1004	Bala	Computer Science	61	80	61	78	51	98	331	66.2	66.2	C	Pass	-	8
8	1011	Gopal	Information Technology	55	48	77	37	96	99	313	62.6	62.6	C	Fail	AI	9
9	1002	Abhay	Information Technology	37	60	78	71	67	98	313	62.6	62.6	C	Fail	Python	9
10	1016	Kamal	Computer Science	77	75	63	39	52	100	306	61.2	61.2	C	Fail	AI	11
11	1005	Bhati	Information Technology	71	59	65	51	59	98	305	61.0	61.0	C	Pass	-	12
12	1018	Lalli	AI and Data Science	47	72	42	84	60	100	305	61.0	61.0	C	Pass	-	12
13	1012	JaiKanth	AI and Data Science	48	60	54	84	52	94	298	59.6	59.6	D	Pass	-	14
14	1010	Gerage	Computer Science	58	36	88	41	61	96	284	56.8	56.8	D	Fail	Java	15
15	1009	Francies	AI and Data Science	39	90	65	35	46	100	275	55.0	55.0	D	Fail	Python,AI	16
16	1001	Abhi	Computer Science	41	53	33	95	45	97	267	53.4	53.4	D	Fail	Linux	17
17	1003	Bhaskar	AI and Data Science	34	65	72	57	38	98	266	53.2	53.2	D	Fail	Python,SQL	18



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Student ID	Student Na	Departmer	Python	Java	Linux	AI	SQL	Attendanc	Total	Average	Percentag	Grade	Status	Failed Subj	Rank
2	1017	Lal	Informatic	94	69	71	96	91	100	421	84.2	84.2	A	Pass	-	1
3	1007	Chawla	Computer	97	92	82	72	75	98	418	83.6	83.6	A	Pass	-	2
4	1013	Karthi	Computer	99	76	48	60	98	98	381	76.2	76.2	B	Pass	-	3
5	1015	Kala	AI and Dat	44	66	90	84	93	98	377	75.4	75.4	B	Pass	-	4
6	1006	Chandran	AI and Dat	77	47	90	54	83	98	351	70.2	70.2	B	Pass	-	5
7	1008	Dora	Informatic	93	42	86	51	67	98	339	67.8	67.8	C	Pass	-	6
8	1014	Kandan	Informatic	64	92	85	37	61	96	339	67.8	67.8	C	Fail	AI	6
9	1004	Bala	Computer	61	80	61	78	51	98	331	66.2	66.2	C	Pass	-	8
10	1011	Gopal	Informatic	55	48	77	37	96	99	313	62.6	62.6	C	Fail	AI	9
11	1002	Abhay	Informatic	37	60	78	71	67	98	313	62.6	62.6	C	Fail	Python	9
12	1016	Kamal	Computer	77	75	63	39	52	100	306	61.2	61.2	C	Fail	AI	11
13	1005	Bhati	Informatic	71	59	65	51	59	98	305	61	61	C	Pass	-	12
14	1018	Lalli	AI and Dat	47	72	42	84	60	100	305	61	61	C	Pass	-	12
15	1012	JaiKanth	AI and Dat	48	60	54	84	52	94	298	59.6	59.6	D	Pass	-	14
16	1010	Gerage	Computer	58	36	88	41	61	96	284	56.8	56.8	D	Fail	Java	15
17	1009	Francies	AI and Dat	39	90	65	35	46	100	275	55	55	D	Fail	Python,AI	16
18	1001	Abhi	Computer	41	53	33	95	45	97	267	53.4	53.4	D	Fail	Linux	17
19	1003	Bhaskar	AI and Dat	34	65	72	57	38	98	266	53.2	53.2	D	Fail	Python,SQ	18
20																
21																

Final Output ¶

[86]:

```
import pandas as pd
import numpy as np
from IPython.display import display

# Student data
data = {
    "Student_ID": [
        1001, 1002, 1003, 1004, 1005, 1006,
        1007, 1008, 1009, 1010, 1011, 1012,
        1013, 1014, 1015, 1016, 1017, 1018
    ],

    "Name": [
        "Lal", "Chawla", "Karthi", "Kala", "Chandran",
        "Dora", "Kandan", "Bala", "Gopal", "Abhay",
        "Kamal", "Bhati", "Lalli", "JaiKanth", "Gerage",
        "Francies", "Abhi", "Bhaskar"
    ],

    "Department": [
        "Information Technology", "Computer Science", "Computer Science",
        "AI and Data Science", "AI and Data Science", "Information Technology",
        "Information Technology", "Computer Science", "Information Technology",
        "Information Technology", "Computer Science", "Information Technology",
        "AI and Data Science", "AI and Data Science", "Computer Science",
        "AI and Data Science", "Computer Science", "AI and Data Science"
    ],

    "Python": [94, 97, 99, 44, 77, 93, 64, 61, 55, 37, 77, 71, 47, 48, 58, 39, 41, 34],
    "SQL": [91, 75, 98, 93, 83, 67, 61, 51, 96, 67, 52, 59, 60, 52, 61, 46, 45, 38],
    "Java": [69, 92, 76, 66, 47, 42, 92, 80, 48, 60, 75, 59, 72, 60, 36, 90, 53, 65],
    "Linux": [71, 82, 48, 90, 90, 86, 85, 61, 77, 78, 63, 65, 42, 54, 88, 65, 33, 72],
    "AI": [96, 72, 60, 84, 54, 51, 37, 78, 37, 71, 39, 51, 84, 84, 41, 35, 95, 57],
    "Attendance": [100, 98, 98, 98, 98, 98, 96, 98, 99, 98, 100, 98, 100, 94, 96, 100, 97, 98]
}

# Create DataFrame
df = pd.DataFrame(data)

print("-" * 50)
print("STUDENT PERFORMANCE ANALYTICS SYSTEM")
print("-" * 50)

print("\n---------- STUDENT DATA ----------")
print(df.to_string(index=False))

# Subject columns
subjects = ["Python", "SQL", "Java", "Linux", "AI"]

# Calculate total and average using NumPy
df["Total"] = np.sum(df[subjects].to_numpy(), axis=1)
df["Average"] = np.mean(df[subjects].to_numpy(), axis=1).round(2)

# Grade function
def grade(avg):
    if avg >= 90:
        return "A+"
    elif avg >= 80:
        return "A"
    elif avg >= 70:
        return "B"
    elif avg >= 60:
        return "C"
    else:
        return "D"
```

```python
df["Grade"] = df["Average"].apply(grade)

# Pass / Fail function
def check_result(avg):
    return "Pass" if avg >= 60 else "Fail"

df["Result"] = df["Average"].apply(check_result)

# Display student performance
performance = df[
    ["Student_ID", "Name", "Department",
     "Total", "Average", "Grade", "Attendance", "Result"]
]

print("\n---------- STUDENT PERFORMANCE ----------")
print(performance.to_string(index=False))

# Highest, lowest and overall average
print("\n---------- OVERALL ANALYSIS ----------")
print("Highest Total Marks:", df["Total"].max())
print("Lowest Total Marks:", df["Total"].min())
print("Overall Average Marks:", round(df["Average"].mean(), 2))

# Top 5 students
print("\n---------- TOP 5 STUDENTS ----------")
top_5 = df.nlargest(5, "Total")
print(top_5[["Student_ID", "Name", "Total", "Average", "Grade"]].to_string(index=False))

# Subject-wise average
print("\n---------- SUBJECT-WISE AVERAGE ----------")
for subject in subjects:
    print(subject, ":", round(df[subject].mean(), 2))

# Pass and fail count
print("\n---------- PASS / FAIL SUMMARY ----------")
print("Pass Students:", (df["Result"] == "Pass").sum())
print("Fail Students:", (df["Result"] == "Fail").sum())

# ---------- FAILED STUDENTS ----------
print("\n---------- FAILED STUDENTS ----------")
failed_students = df[df["Result"] == "Fail"]

if failed_students.empty:
    print("No students failed.")
else:
    display(
        failed_students[
            ["Student_ID", "Name", "Department", "Total", "Average", "Grade", "Attendance", "Result"]
        ].reset_index(drop=True)
    )
```

```
--------------------------------------------------
STUDENT PERFORMANCE ANALYTICS SYSTEM
--------------------------------------------------

---------- STUDENT DATA ----------
 Student_ID     Name             Department  Python  SQL  Java  Linux  AI  Attendance
       1001      Lal Information Technology      94   91    69     71  96         100
       1002   Chawla       Computer Science      97   75    92     82  72          98
       1003   Karthi       Computer Science      99   98    76     48  60          98
       1004     Kala    AI and Data Science      44   93    66     90  84          98
       1005 Chandran    AI and Data Science      77   83    47     90  54          98
       1006     Dora Information Technology      93   67    42     86  51          98
       1007   Kandan Information Technology      64   61    92     85  37          96
       1008     Bala       Computer Science      61   51    80     61  78          98
       1009    Gopal Information Technology      55   96    48     77  37          99
       1010    Abhay Information Technology      37   67    60     78  71          98
       1011    Kamal       Computer Science      77   52    75     63  39         100
       1012    Bhati Information Technology      71   59    59     65  51          98
       1013    Lalli    AI and Data Science      47   60    72     42  84         100
       1014 JaiKanth    AI and Data Science      48   52    60     54  84          94
       1015   Gerage       Computer Science      58   61    36     88  41          96
       1016 Francies    AI and Data Science      39   46    90     65  35         100
       1017     Abhi       Computer Science      41   45    53     33  95          97
       1018  Bhaskar    AI and Data Science      34   38    65     72  57          98

---------- STUDENT PERFORMANCE ----------
 Student_ID     Name             Department  Total  Average Grade  Attendance Result
       1001      Lal Information Technology    421     84.2     A         100   Pass
       1002   Chawla       Computer Science    418     83.6     A          98   Pass
       1003   Karthi       Computer Science    381     76.2     B          98   Pass
       1004     Kala    AI and Data Science    377     75.4     B          98   Pass
       1005 Chandran    AI and Data Science    351     70.2     B          98   Pass
       1006     Dora Information Technology    339     67.8     C          98   Pass
       1007   Kandan Information Technology    339     67.8     C          96   Pass
       1008     Bala       Computer Science    331     66.2     C          98   Pass
       1009    Gopal Information Technology    313     62.6     C          99   Pass
       1010    Abhay Information Technology    313     62.6     C          98   Pass
       1011    Kamal       Computer Science    306     61.2     C         100   Pass
       1012    Bhati Information Technology    305     61.0     C          98   Pass
       1013    Lalli    AI and Data Science    305     61.0     C         100   Pass
       1014 JaiKanth    AI and Data Science    298     59.6     D          94   Fail
       1015   Gerage       Computer Science    284     56.8     D          96   Fail
       1016 Francies    AI and Data Science    275     55.0     D         100   Fail
       1017     Abhi       Computer Science    267     53.4     D          97   Fail
       1018  Bhaskar    AI and Data Science    266     53.2     D          98   Fail

---------- OVERALL ANALYSIS ----------
Highest Total Marks: 421
Lowest Total Marks: 266
Overall Average Marks: 65.43

---------- TOP 5 STUDENTS ----------
 Student_ID     Name  Total  Average Grade
       1001      Lal    421     84.2     A
       1002   Chawla    418     83.6     A
       1003   Karthi    381     76.2     B
       1004     Kala    377     75.4     B
       1005 Chandran    351     70.2     B

---------- SUBJECT-WISE AVERAGE ----------
Python : 63.11
SQL : 66.39
Java : 65.67
Linux : 69.44
AI : 62.56

---------- PASS / FAIL SUMMARY ----------
Pass Students: 13
Fail Students: 5
```

---------- FAILED STUDENTS ----------

	Student_ID	Name	Department	Total	Average	Grade	Attendance	Result
0	1014	JaiKanth	AI and Data Science	298	59.6	D	94	Fail
1	1015	Gerage	Computer Science	284	56.8	D	96	Fail
2	1016	Francies	AI and Data Science	275	55.0	D	100	Fail
3	1017	Abhi	Computer Science	267	53.4	D	97	Fail
4	1018	Bhaskar	AI and Data Science	266	53.2	D	98	Fail